

AI MUSIC PRODUCTION PLATFORM

FINAL INSTALLATION COMPLETION EXECUTION PLAN

מטרת המשימה

המטרה היא לסיים את כל שכבת ההתקנות, provisioning, attestation, deployment וה-API connection של המערכת לפני שעוברים לשלב המוזיקלי.

בסיום לא יהיו providers במצב מעורפל.

כל provider חייב להיות בדיוק באחד המצבים:

READY
RESEARCH_READY
BLOCKED_LICENSE
BLOCKED_NO_WEIGHTS
BLOCKED_UPSTREAM
BLOCKED_MISSING_LICENSED_ASSET

אין להשתמש ב:

configured
almost ready
mostly installed
scaffolded
pending

כ-final status.

1. SOURCE OF TRUTH

הקובץ הבא כבר קיים:

installation-matrix-v2.json

הוא authority היחיד על final status.

גם קיימים:

scripts/audit-installation-stack.py
docs/installation-completion-report.md

יש להשתמש בהם לאורך כל העבודה.

אחרי כל provider שמסתיים:

1. לעדכן "installation-status.json" provider-local.
2. לעדכן "installation-matrix-v2.json".
3. לעדכן "docs/installation-completion-report.md".
4. להריץ:

python scripts/audit-installation-stack.py

אסור לשנות "READY" ידנית רק כדי שה-audit יעבור.

2. לא לגעת במה שכבר נסגר

BEAT_THIS

מצבו הנוכחי:

READY

יש:

- immutable source
- checkpoint SHA
- L4 runtime
- real audio smoke
- beats 92
- downbeats 16
- signed promotion
- endpoint
- Node attestation
- live health

אין לבצע reinstall.

רק health regression check בסוף כל wave.

MR_MT3

מצבו:

READY

יש:

- immutable model revision

checkpoint SHA256 -
note CUDA smoke-63 -
signed promotion -
live endpoint -
health ready -

אין לבצע reinstall או model replacement.

SHEETSAGE

מצבו:

RESEARCH_READY

יש כבר:

exact source -
exact wheel hash -
exact asset manifest -
production deployment -
persistent signed real smoke -
melody events 21 -
chord events 4 -
timing events 35 -
live Node path -

להשאיר Research-only עקב הרישוי.

לא להמשיך כרגע בעוד hardening של recovery/capacity אלא אם קיים bug אמיתי.

3. PHASE A — REPAIR PROVIDERS שפועלים אבל נכשלים ב-AUDIT

זה הדבר הראשון לבצע.

לא להתחיל provider חדש לפני שסוגרים את אלה.

3.1 ACE_STEP — FIX IDENTITY MISMATCH

כרגע יש:

real smoke = yes
deployment = yes

promotion signature = valid

אבל:

live sourceRevision != signed promotion sourceRevision

ולכן הוא כרגע אינו READY audit-wise.

לעשות:

A. לקרוא את "/health" live

לאסוף:

provider
modelVersion
checkpointSha256
checkpointRevision
sourceRevision
sourceImageDigest
modalAppld
modalDeploymentId
modalFunctionId
modalImageId
runtime

B. להשוות מול promotion bundle

אם deployment הנוכחי הוא הנכון:

יש לבצע promotion חדש על ה-deployment החדש הקיים.

אם deployment הנוכחי אינו מה-source המאושר:

יש לבצע redeploy מתוך ה-commit המאושר.

אסור:

edit promotion to match live endpoint

בלי provenance אמיתי.

C. Real smoke מחדש

יש להריץ generation אמיתי.

נדרש:

realInference=true
nonSilent=true
valid WAV
checkpoint identity exact

D. Sign promotion

חתום על identity המדויק.

E. Verify API attestation

ה-API Node חייב לדחות:

source mismatch -
checkpoint mismatch -
image mismatch -
.invalid signature -

:Success

ACE_STEP = READY

ALL_IN_ONE — FIX PROMOTION SIGNATURE 3.2

כרגע ה-live health תקין.

הבעיה:

retained promotion signature invalid under current public key

לעשות:

1. לקרוא live identity.
2. לוודא:
 - model version
 - all eight Harmonix folds
 - HTDemucs SHA
 - image
 - source revision
3. אם live deployment נכון:
 - ליצור promotion חדש עם signing key הנוכחי.
4. אם public key הוא זה שהוחלף בטעות:
 - לא להחזיר key ישן בלי לוודא provenance.
5. להריץ real structure smoke מחדש.
6. verify API signature validation.
7. לעדכן matrix.

:Success

ALL_IN_ONE = RESEARCH_READY

אם weights נשארים NC.

לא לסמן commercial READY.

MT3 BASELINE — FIX PROMOTION 3.3

אותה בעיה בסיסית:

deployment exists
promotion signature not valid against current configured key

לעשות:

1. verify exact checkpoint SHA

...33f6bc4c

2. verify exact model conversion provenance

3. real transcription smoke

4. require actual terminated MIDI notes

5. create fresh signed promotion

6. verify API

7. persist health evidence

:Final

MT3 = RESEARCH_READY

או READY רק אם model/checkpoint licensing really supports it

BS_ROFORMER — RESTORE FULL EVIDENCE CHAIN 3.4

הוא כנראה עובד בפועל, אבל ה-root audit לא מקבל אותו כ-READY כי חסרה שרשרת retained evidence מלאה.

לעשות:

אין להחליף checkpoint.

יש לשחזר evidence עבור:

source revision
checkpoint SHA
config SHA
runtime
license
real separation smoke
deployment identity
endpoint
API routing
promotion
live health

:Real smoke

→ שיר אמיתי

vocals
instrumental
bass / available stems

חובה לוודא:

not silent
valid duration
stems not identical
SHA persisted

:Final

BS_ROFORMER = READY

DEMUCS 3.5

יש live health ו-checkpoint hash, אבל ה-audit דורש full retained chain.

יש להשלים:

source version/revision
package pin
license
checkpoint identity
real separation smoke
API path evidence

לא צריך promotion אם policy אינה דורשת.

:Final

DEMUCS = READY

BASIC_PITCH 3.6

אותה לוגיקה.

יש package/checkpoint health

להוסיף:

source identity

license

real transcription fixture

actual MIDI notes

API path

retained status

:Final

BASIC_PITCH = READY

MIR LEGACY STACK 3.7

לסגור evidence עבור:

MADMOM

ESSENTIA

CHROMA

TORCHCREPE

PYLOUDNORM

כרגע הם חסומים בעיקר בגלל lack of retained chain.

לכל אחד:

MADMOM

.real beat/downbeat inference

ESSENTIA

.real tonal/key inference

CHROMA

.real HPCP/chroma output

TORCHCREPE

.stem על real pitch F0 curve

PYLOUDNORM

.real LUFS measurement

לכל provider לשמור:

package

version

runtime

license

real fixture SHA

output evidence

endpoint identity

health

Node path

לא לבנות אותם מחדש אלא אם ה־runtime חסר.

PHASE B — COMPLETE ANALYSIS STACK .4

SONGFORMER 4.1

כרגע:

BLOCKED_NO_WEIGHTS

יש source/model manifest אבל אין provisioned assets.

לעשות:

Provision exact model .1

להוריד exact pinned HF snapshot.

Dependencies .2

:Provision

SongFormer

MusicFM

MuQ dependencies required by SongFormer

config files

stats

SHA inventory .3

לכל קובץ:

path

bytes

sha256

source

revision

license

אין להסתפק ב־MD5 upstream.

Runtime build .4

לנעול requirements.

Real full-song smoke .5

Output חייב לכלול:

boundaries

labels

confidence

לא output ריק.

Modal deploy .6

.Deploy isolated GPU worker

:Add .7

SONGFORMER_API_URL

Node wiring .8

ה־Node analysis provider חייב לקבל:

sections
labels
timestamps
confidence
provenance

Health .9

:Require

runtimeReady
assetsVerified
smokePassed
modelRevision
assetManifestSha

:Final

SONGFORMER = READY / RESEARCH_READY according to exact weight license

YOUR_MT3 4.2

כרגע נכשל אמיתי:

GPU smoke = 0 notes

אסור פשוט להשתמש ב־fixture אחר עד שיוצא משהו.

לעשות:

קודם לפתור provenance

להשיג immutable model revision.

:Compute

checkpoint SHA256

ואז להבין למה 0 notes

לבדוק:

sample rate
mono/stereo
normalization
input duration

frontend
model config
checkpoint compatibility
tokenizer
inference thresholds
postprocess
compatibility patch

להשתמש ב־fixture ברור

לא silence/click-only בלבד.

:Create deterministic musical smoke

piano notes -
sustained chords -
known pitches -
.sufficient duration -

:Required

terminated MIDI note 1 =<

רצוי הרבה יותר.

:Then

signed promotion -
endpoint -
health -
.Node parse -

:Final

YOUR_MT3 = READY

:exact upstream environment אם לא מצליח אחרי

BLOCKED_UPSTREAM

בלי hack.

PHASE C — MOSS-MUSIC .5

זה priority גבוה.

כרגע:

BLOCKED_UPSTREAM

בגלל conflict בין:

TorchCodec 0.8

TorchCodec 0.9

אין לעשות:

pip --force

no-deps blindly--

patch package metadata only

לעשות:

Build compatibility matrix .1

לבדוק combinations של:

MOSS-Music commit

MOSS-SGLang moss-audio commit

Torch version

TorchCodec version

FFmpeg version

.2. לבחור pair רשמי/תואם

לא "latest".

.3. Require:

pip check

.PASS

.4. Media test

decode WAV

decode MP3

resample

audio tensor creation

.5. Instruct model

.Real song

:Structured prompt

tempo
key
sections
instrumentation
style
arrangement observations

Thinking model .6

אותו שיר.

.Require meaningful long-form reasoning

Persist model hashes .7

Instruct ו־Thinking בנפרד.

Deploy .8

:Canonical endpoint

MOSS_MUSIC_API_URL

אפשר worker משותף עם model selector אם GPU memory מאפשר, אחרת שני deployments.

Never make MOSS canonical MIR truth .9

:Output category

SEMANTIC_MUSIC_REASONING

:Final

MOSS_MUSIC_INSTRUCT = READY
MOSS_MUSIC_THINKING = READY

או "BLOCKED_UPSTREAM" אם אין coherent runtime.

PHASE D — VOCAL-TO-ACCOMPANIMENT FAMILY .6

HAFM 6.1

זה כנראה הבא הכי קל לסגור.

כבר provisioned מספר model files.

ה-blocker בפועל:

real-vocal.wav missing

ליצור fixture מורשה

השתמש בהקלטת ווקאל מקורית בבעלותך.

לא commercial song extracted stem אם הזכויות אינן ברורות.

:Save outside Git

var/lib/hafm/smoke/real-vocal.wav/

:Record

fixture SHA

license/ownership note

duration

sample rate

Run real HAFM inference

:Require

instrumental WAV

non-silent

finite

duration plausible

not source copy

:Then

persist asset tree SHA -

promotion -

L40S endpoint -

"HAFM_API_URL" -

health -

.API connection -

:Final

HAFM = READY

LADA_BAND 6.2

:Current

BLOCKED_LICENSE

המודל מחקרי/gated.

אם private non-commercial research allowed:

1. record explicit license acceptance.

2. obtain gated HF assets.

3. download entire required:

/checkpoints

/pretrained

4. hash everything.

5. run real dry vocal.

6. verify real accompaniment.

7. deploy research endpoint.

8. keep excluded from commercial router.

:Final

LADA_BAND = RESEARCH_READY

לא READY.

אם access/license לא ניתן:

BLOCKED_LICENSE

ANYACCOMP 6.3

הבעיה הראשונה:

source revision missing

weights revision missing

לעשות:

1. Resolve exact Amphion source commit.

2. Resolve exact HF "amphion/anyaccomp" commit.

3. Update manifest.

4. Download.

VQ -
Flow Matching -
Vocoder -
configs -
.hash all .5
.build exact Python 3.9/Torch 2.3.1 image .6
.run real vocal .7
.require generated accompaniment .8
.deploy endpoint .9
:set .10

ANYACCOMP_API_URL

.License review mandatory

:If research only

RESEARCH_READY

MIDI-SAG + MUSECONTROLLITE 6.4

ה-worker קיים אבל asset provenance חסר.

יש להשלים אחד-אחד:

RMVPE
GAME-1.0-medium
MuseControlLite
Vocal Beat Tracking
AccoMontage2
CSLL2M
SoulX-Singer
SoulX-Singer-Preprocess

לכל asset:

exact source
revision
URL
filename
bytes
SHA256
license

Smoke 1

:MuseControlLite standalone

melody
chroma
rhythm
dynamics
audio →

Smoke 2

:Full MIDI-SAG detected mode

vocal
beat →
MIDI →
chords →
neural accompaniment →

:Persist intermediate

beat map
MIDI SHA
chord map
final audio SHA

Deploy

MIDI_SAG_API_URL

:Expose

MUSE_CONTROL_LITE

.as separate capability

:Final

.license dependent

PHASE E — GENERATION / REPAIR ENGINES .7

MUSICGEN LARGE + MELODY LARGE 7.1

ה-resolved immutable HF revisions כבר

עכשיו צריך provisioning אמיתי.

:Provision

facebook/musicgen-large
facebook/musicgen-melody-large

.exact revisions only

:Build exact runtime

Python 3.9
CUDA 11.8
Torch 2.1
FFmpeg 4.x
AudioCraft pinned commit

Smoke Large

.Text prompt → WAV

Smoke Melody

.Reference melody → WAV

:Require

non-silent
not source copy
correct provider identity

Deploy

MUSICGEN_API_URL

:Final

RESEARCH_READY

.weights רישיון

JASCO 7.2

כרגע:

BLOCKED_NO_WEIGHTS

אל תעשה שום workaround.

רק אם נמצא:

immutable checkpoint
revision
SHA
license

אז install בתוך AudioCraft worker.

אחרת להשאיר:

BLOCKED_NO_WEIGHTS

DIFFRHYTHM2 7.3

כבר נעשה GPU attempt אבל לא יצא artifact.

צריך לבצע debugging אמיתי.

לבדוק:

checkpoint presence
MuQ-MuLan assets
tokenizer
reference preprocessing
prompt/lyrics format
duration
GPU OOM
output path
VAE/codec
dtype
CUDA compatibility

First target

.Generate 10–20 sec known-good output

רק אחרי שזה עובד, run full real fixture.

:Require

artifact exists

valid audio
non-silent
not source-copy
lyricsConditioned
rhythmConditioned

:Then

persistent smoke -
promotion -
endpoint -
."DIFFRHYTHM2_API_URL" -

:Final

DIFFRHYTHM_2 = READY

STABLE AUDIO 3 7.4

כרגע המודלים pinned אבל לא provisioned.

קודם License

.Store explicit Stability license acceptance

:Provision

stable-audio-3-small-music
stable-audio-3-medium

.exact existing revisions

Build deploy layer

:Ensure presence of

Dockerfile
modal_app.py
modal_config.py
modal_provision.py
smoke.py

Four smoke types

A. Text-to-audio

B. Audio-to-audio

C. Continuation

D. Inpainting

.Inpainting smoke is mandatory

:Known WAV

mask exact region
generate replacement →

:Verify

inside region changed
outside region preserved

:Deploy

STABLE_AUDIO_3_API_URL

:Final

STABLE_AUDIO_3_SMALL_MUSIC = READY
STABLE_AUDIO_3_MEDIUM = READY

.subject to license status

PHASE F — CLaMP3 .8

.Worker scaffold already exists

:Current blocker

5GB weights not provisioned~

לעשות:

.Provision all exact pinned assets .1

.Hash complete tree .2

.GPU build .3

.real semantic smoke .4

:Required tests

matching audio/text score > mismatched
matching MIDI/audio score > mismatched

.deploy endpoint .5
:configure .6

CLAMP3_API_URL

.Node route already exists — verify actual live path .7

:Final

CLAMP3 = READY / RESEARCH_READY according to weights license

PHASE G — OPEN VOCABULARY SEPARATION .9

SAM AUDIO

.blocked on license / no installation כרגע

:First action

.Perform exact license review

.Do not assume commercial availability

:If current private research use permitted

Create worker

/services/sam-audio-worker

Pin source

.Exact commit

:Provision

:Prefer

sam-audio-base
sam-audio-large

Smoke

.Real mixed song

:Prompts

vocals
acoustic guitar
drums
strings

:Require

stem audible
residual audible
stem != residual
valid duration

:Deploy

SAM_AUDIO_API_URL

:Final

READY

:or

RESEARCH_READY

.according to exact license

PHASE H — MAGENTA REALTIME 2 .10

:Install

MAGENTA_RT2_SMALL
MAGENTA_RT2_BASE

:Assets

MusicCoCa
SpectroStream
small
base

:Runtime

.Use pinned JAX/CUDA stack

.No floating package versions

Smoke 1

.Text-conditioned audio

Smoke 2

.MIDI-conditioned

.Known MIDI sequence must alter output

Smoke 3

.Audio-context continuation

Important

:Provider metadata must state

GENERATIVE_MIDI_CONDITIONED

.not deterministic renderer

:Deploy

MAGENTA_RT2_API_URL

:Final

MAGENTA_RT2_SMALL = READY
MAGENTA_RT2_BASE = READY

PHASE I — SYMBOLIC ARRANGEMENT STACK .11

REMI-Z 11.1

.Install actual package/source

.Pin exact revision

:Smoke

MIDI → REMI-z → MIDI

.Roundtrip must preserve valid musical structure

:Final

REMI_Z = READY / RESEARCH_READY

MUSIC2MUSIC 11.2

.License first

:If usage allowed

:Provision

m2m_arranger
m2m_pianist_dur
m2m_drummer
m2m_pt

.Exact revision and SHA

:Smoke

known MIDI
multitrack rearrangement →

:Require

multiple tracks
actual notes
valid MIDI
output differs from input

:Final

RESEARCH_READY

.until commercial license is established

METEOR 11.3

:Before install

.resolve licensing

:If allowed

Python 3.9.2 -
exact source -
exact checkpoint -
preprocessing pipeline -
.REMI/MuseMorphose assets -

:Smoke

MIDI → tokenization → generation → MIDI

:Final

.license-dependent

SYMPHONYGEN 11.4

.Resolve source license before production

:Provision

stage_one_pretrained.pt
stage_two_pretrained.pt
grpo_clamp_epoch_10.pt
grpo_clamp+track_epoch_6.pt

.SHA all four

:Smoke

symbolic input → orchestral multitrack MIDI

.Require real notes across multiple tracks

PHASE J — EVALUATION STACK .12

MuQ 12.1

.Install shared embedding service

.Resolve exact license

.Provision model

:Smoke

same-song segment similarity

<

unrelated-song similarity

MuQ-MuLan 12.2

.Text/audio similarity

:Smoke

correct description

<

wrong description

MuQ-Eval 12.3

.Provision exact A1 checkpoint

:Smoke

clean fixture -

.deliberately degraded fixture -

:Expected

clean > degraded

.Do not hardcode threshold

:Deploy

MUQ_EVAL_API_URL

Song Aesthetics 12.4

.Install exact model/checkpoint

:Input

mixture
vocal
accompaniment

:Output

Overall
Singing
Melody
Arrangement
AudioQuality

.Real inference required

SongEval 12.5

.Research only unless license changes

.Provision model and run real song

:Final

RESEARCH_READY

PHASE K — NATIVE RENDERERS .13

SFIZZ + VSCO2 CE 13.1

.installation כרגע יש bootstrap code אבל לא

.Pinned source already identified

Build sfizz 1.2.3

.Create native executable

:Record

binary SHA
source commit
compiler/runtime

Provision VSCO2 CE

:Private asset directory

var/lib/music-ai/assets/sfz/vsco2-ce/

.Calculate tree SHA

Approve host

:Add exact host identity to

MUSIC_AI_APPROVED_NATIVE_HOSTS

Run 3 real TrackModel renders

pitch variant .1
velocity/expression variant .2
articulation/CC variant .3

.Require all three WAV hashes different

:Final

SFIZZ_VSCO2_CE = READY

VST3 HOST 13.2

.Build actual native host

:Must consume

notes
velocity
CC
pitch bend
automation
articulation

.Produce offline WAV

.Host must have exact SHA and approved identity

:If no VST3 instrument supplied

VST3_HOST = READY
VST3_INSTRUMENT = BLOCKED_MISSING_LICENSED_ASSET

.Do not install random plugin

PHASE L — LEVO2 RESEARCH .14

.Low priority

.Only after all above

:If license permits your private research

.Provision exact weights

.Run real full-song generation

.Keep excluded from commercial routing

:Final

LEVO2_RESEARCH = RESEARCH_READY

:Otherwise

BLOCKED_LICENSE

WATCHLIST — NO WORK NOW .15

:Keep only metadata for

S2Accompanist

BeatEdit

Live Music Diffusion Models

Diff-A-Riff

Break-the-Beat

.Do not create workers until installable public checkpoints exist

EXECUTION ORDER — DO NOT CHANGE .16

Wave 1 — Repair existing production foundation

ACE_STEP
ALL_IN_ONE
MT3
BS_ROFORMER
DEMUCS
BASIC_PITCH
MADMOM
ESSENTIA
CHROMA
TORCHCREPE
PYLOUDNORM

Wave 2 — Complete analysis/transcription

SONGFORMER
YOUR_MT3
MOSS_MUSIC

Wave 3 — V2A

HAFM
LADA_BAND
ANYACCOMP
MIDI_SAG
MUSE_CONTROL_LITE

Wave 4 — Generators/repair

MUSICGEN
JASCO status
DIFFRHATHM2
STABLE_AUDIO_3
CLAMP3

Wave 5 — New capabilities

SAM_AUDIO
MAGENTA_RT2

Wave 6 — Symbolic

REMI_Z
MUSIC2MUSIC

METEOR
SYMPHONYGEN

Wave 7 — Evaluation

MUQ
MUQ_MULAN
MUQ_EVAL
SONG_AESTHETICS
SONG_EVAL

Wave 8 — Rendering

SFIZZ
VSCO2
VST3_HOST
VST3_INSTRUMENT_STATUS

Wave 9 — Optional research

LEVO2

AFTER EVERY WAVE .17

:Run

python scripts/audit-installation-stack.py

.Then all provider-specific tests

.Then API tests

.Then update completion report

Do NOT move to next Wave when a provider from the current Wave is merely “almost
.”ready

:Either close it as

READY
RESEARCH_READY

.or record truthful blocker

DO NOT REPEAT OVERENGINEERING .18

:Do not spend additional cycles now on

- SheetSage recovery alert refinements
- new capacity dashboards
- new deployment abstractions
- generic provider frameworks
- extra orchestration layers
- new model discovery

.unless required to close a specific blocker

.The project currently needs completed installations, not more infrastructure

FINAL ACCEPTANCE .19

:The installation phase is complete only when

python scripts/audit-installation-stack.py

:returns

PASS

.and every provider in "installation-matrix-v2.json" has a truthful terminal status

:Additionally

Zero "null" immutable revisions

.except truly non-versioned external licensed assets with SHA identity

Zero request-time model downloads

:Zero READY providers with

- smoke=false
- health=false
- endpoint=false
- assetsChecksummed=false

:Zero production READY providers with

license=RESEARCH_ONLY

Zero promotion mismatches

Zero invalid signatures

Zero fake/synthetic inference results

STOP .20

:When the installation audit is fully closed

.STOP

.Do not start Golden Song automatically

.Do not improve the Arrangement Director

.Do not generate arrangement candidates

:The next explicit phase will be

REAL MUSICAL CERTIFICATION

:which will include

provider shootouts

Golden Song

analysis accuracy

SongModel fusion

ACE-Step vs LaDA vs HAFM vs AnyAccomp

repair comparison

symbolic arranger comparison

evaluation ensemble

human listening

Arrangement Director V2

professional-quality target

FINAL PRINCIPLE

:For every provider

CODE

EXACT REVISION →

EXACT WEIGHTS →

SHA256 →

LICENSE →
ISOLATED RUNTIME →
REAL INFERENCE →
RETAINED PROOF →
ATTESTATION →
DEPLOYMENT →
API CONNECTION →
LIVE HEALTH →
READY →

.If one link is missing, the provider is not READY

.The objective of this phase is not to make the repository look bigger

.The objective is to make every installed capability real, reproducible, auditable and usable